KYLE NERO A. SANTOS



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PROFESSIONAL SUMMARY

Electronics Engineering graduate specializing in embedded systems, computer vision, and software development. Experienced in designing hardware-software solutions using ESP32 and YOLO-based object detection, alongside developing cross-platform applications and enterprise workflow automation.

EDUCATION

Bachelor of Science in Electronics Engineering
June 2022 - Present

Marikina Polytechnic College

Capstone: Development of an Automated Optical Detector for Microplastics in Rock Salt for Retail and Domestic Use

Achievements: 1st Placer in Engineering Capstone Project Exhibit (2026), Participated in Research Innovation Congress (2026)

INTERNSHIP

First Option Electronic System Phils., Inc.
June 2025 - Aug 2025

Pasig City Metro Manila - Philippines

- Managed end-to-end surveillance operations from designing comprehensive CCTV layouts to overseeing control room personnel, enhancing overall site security and operational efficiency.

TECHNICAL SKILLS

Programming: Python, OpenCV, C/C++, Arduino/ESP32 firmware logic, TypeScript, React Native, SQL, JavaScript

ITSM Automation: ServiceNow Basic Fundamentals, ServiceNow Catalog (Including and not limited to Creation of Catalog Item, Basic Flow Designer, And System Administration/Navigation)

AI and Computer Vision: model training, AI Automation, object detection, dataset annotation

API and Data Handling: API concepts, Supabase, integrating AI/API-backed features, Database Management, JSON

Tools and Simulation: VS Code, Antigravity IDE, Claude Code, Arduino IDE, Git/GitHub, Excel, MS Office

ENGINEERING EDUCATIONAL PROJECT EXPERIENCE

Capstone Project | MICROTECT: Automated Microplastic Detector
April 2026

- Led the end-to-end system development of a Raspberry Pi 5 optical detection prototype, eliminating the time and cost limits of manual rock salt microscopy.
- Designed the complete software architecture, implementing a YOLOv11 and OpenCV detection workflow that achieved 0.826 precision, 0.733 recall, and 0.814 mAP@50.
- Developed an end-to-end operator interface, automating the generation of microplastic contamination and particle count reports.
- Executed controlled multi-trial testing across diverse rock salt samples, validating prototype performance under varied conditions.

Software Applications Project | HerCare and KKB Snap
Product and Frontend Development

- Developed interactive mobile product flows for a women's health companion app (HerCare) using React, TypeScript, and Vite, delivering a seamless, user-focused interface.
- Developed a cross-platform receipt-splitting application using TypeScript, React Native, Expo, Supabase PostgreSQL, Realtime, and Mindee OCR, enabling receipt scanning, live item claiming, proportional bill calculation, and payment QR handoff.

CERTIFICATIONS

ServiceNow Micro-Certification - Flow

ServiceNow | June 2026

- Completed official certification in designing, testing, and deploying automation workflows and managing data payloads.

ServiceNow Administration Fundamentals — **In Progress**

- Pursuing comprehensive platform administration training, preparing for the official Certified System Administrator (CSA) credential.
- Completed foundational system administrator microlearning modules, establishing a practical baseline for enterprise workflow management.

AFFILIATIONS

- Institute of Electronics Engineers of the Philippines, Student Member